NAME: ______________________ **PERIOD:** ________ **DATE:** ______________

One Result, Two Claims

AP Statistics · Unit 3 · Topic 3.13 · B: 2027-01-05 · E: 2027-01-08

[STAR] TODAY'S PROBLEM

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, randomly assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: "I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A."

Salma: "Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement."

Conditions have been verified: random selection and assignment; $10{,}000 \leq 50{,}000$; pooled expected counts 3,000 and 2,000 in each group.

On-time payment

Text	n	Paid	Other
A	5000	3050	1950
B	5000	2950	2050

FIRST TAKE — before we discuss (5 min)

The observed improvement is 2 percentage points and the deployment target is 5. Is that enough to deploy? Give one reason.

[BOOK] CALCULATE, INTERPRET, THEN BOUND THE CLAIM (keep on the page)

For $H_0 : p_1 - p_2 = 0$, pool $\hat{p}_c = (x_1 + x_2)/(n_1 + n_2)$ and use $z = \dfrac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1-\hat{p}_c)(1/n_1 + 1/n_2)}}$. A two-sided test uses $P(|Z| \geq |z|)$. The p-value assumes H_0 is true and measures how often random chance would produce a difference at least as extreme. If the p-value is at most α, reject H_0; otherwise fail to reject it. A test against zero addresses whether a difference exists; it does not by itself establish that the difference exceeds a separate practical threshold.

Work (a)–(c) from the rules box:

DOK 1 **(a)** Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

DOK 2 **(b)** Using the verified conditions, calculate the pooled proportion, pooled SE, z, and the two-sided p-value for $H_0 : p_A - p_B = 0$.

DOK 3 **(c)** In 3–4 sentences, choose a report using the p-value, the 2-point observed gap, and the 5-point target. Explain what the test against zero establishes and why that does or does not justify deployment.

Frame: The p-value supports ____________________, because ____________________.

Frame: The test compares the difference with __________, but deployment requires ______________; therefore ____________________.

EXIT — turn this in

One thing the rules box changed about my first take: ______________________________